

# The Gate, Not the Network?

## A Matched-Control and Endpoint-Metric Audit of Learned Prefetching

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**Abstract**—Learned cache prefetchers are often compared with classical predictors that always issue requests. This confounds address prediction with admission: a learned system may appear safer because it declines to prefetch. We audit an online MLP delta predictor using classical controls with matched admission, then test the mechanism in native ChampSim.

The broad neural interpretation does not survive. On random traffic the MLP and matched classical control are statistically indistinguishable, while classical controls are substantially faster on several regular streams. A simple dominant-delta admission gate nevertheless looks promising in the local model. We therefore freeze a native holdout before observation: the highest-weight public DPC-3 SimPoint for each of the nine remaining SPEC CPU2017 programs, with 50 million warm-up and 200 million measured instructions. The gate-disabled port matches official `ip_stride` exactly. On holdout, gating removes 36.77% of accepted L1D prefetches and raises accuracy from 10.64% to 15.62%, but reduces LLC prefetch misses by only 6.85%, changes DRAM reads by  $-0.083\%$ , and loses 0.566% geometric-mean IPC. Direction is heterogeneous (four wins, five losses; sign-test  $p = 1$ ). Better proxy metrics therefore establish neither lower off-chip traffic nor higher performance. The defensible contribution is a reproducible attribution and endpoint-metric audit, not a neural, energy, or deployment claim.

**Index Terms**—cache prefetching, confidence, matched controls, negative results, reproducibility, ChampSim

### I. INTRODUCTION

Prefetching trades latency for bandwidth and cache capacity [1], [2]. Confidence mechanisms can suppress requests on unpredictable streams, but they complicate attribution. Comparing a gated learned predictor with an always-on stride predictor changes both address prediction and admission. Any apparent neural advantage may instead be a gate advantage.

This work began as a positive study of a 6–32–1 online MLP that predicts the next address delta. An audit found two decisive confounds: the largest speedups assumed zero-latency prefetch completion, and the random-stream result lacked a classical predictor with the same gate. We ask:

What remains after prediction, admission, latency, contention, lookahead, cache organization, experimental unit, and endpoint metric are separated?

Our contributions are:

- matched controls that isolate the MLP from its admission policy;

- a latency- and contention-aware local simulator with a demand-only shadow cache and explicit lateness, pollution, and drop accounting;
- a compact, predictor-independent dominant-delta gate, tuned on local development seeds and evaluated on disjoint local holdout seeds;
- a native runtime-module port whose disabled form is output-identical to official ChampSim `ip_stride`; and
- a frozen nine-program native holdout showing that fewer accepted prefetches and higher accuracy do not imply less DRAM traffic or more IPC.

The first result is an attribution failure: the evidence does not support a general neural advantage. The second is a metric failure: the admission gate improves common proxy metrics without improving the memory-system endpoint.

### II. LOCAL AUDIT MODEL

#### A. Machine and measurement

`nncpu` executes an in-order scalar instruction stream. Fetch costs one cycle; an L1 hit costs one and a demand miss has 40-cycle latency. The L1 has 32 lines of eight 64-bit words. An ideal, fully associative, zero-prefetch-latency profile is retained only as an upper-bound control.

Timed profiles use an 8-way L1 and asynchronous 40-cycle prefetches. Demand that catches an in-flight request waits for its remaining latency. A timed profile starts one memory request per cycle with 16 prefetch MSHRs; a constrained profile starts one every four cycles with eight. Demand and prefetch share the issue schedule. Excess prefetches are dropped.

A same-geometry shadow L1 receives demand accesses only. Comparing real and shadow outcomes identifies useful hits, pollution misses, late prefetches, redundant requests, and MSHR drops. This model improves attribution but is not an out-of-order microarchitectural simulator.

#### B. Predictors and matched admission

Global stride predicts the previous delta. A bounded delta-Markov control records which delta most often follows the current delta. The learned model is an online 6–32–1 ReLU MLP trained from a 64-sample replay buffer. It receives opcode, static PC when present, address, two deltas, and a bias; its target is clamped to  $[-16, 16]$  words. A predictor-independent

lookahead adapter keeps prediction distance separate from admission.

The original prediction-error gate admits when median recent error is at most half the characteristic delta magnitude over 48 accesses. We apply the same gate to stride and delta-Markov, and wrapped predictors continue training while closed. This is the matched control missing from the original claim.

Prediction error can reject repeated multimodal streams. We therefore define recent regularity

$$R_t = \frac{\max_d |\{i : d_i = d\}|}{W}. \quad (1)$$

After eight deltas, the gate admits iff  $R_t \geq 0.35$  for  $W = 16$ . Window and threshold were chosen on development seeds 0–9; local confirmatory numbers use seeds 10–39.

An exact incremental implementation with a 16-entry signed 64-bit FIFO and a 16-entry associative histogram requires 2,153 state bits (269.1 bytes, 0.41% of the DPC-3 64-KiB budget). This is storage accounting, not area, timing, or energy synthesis; comparator and update logic are not costed.

### III. EXPERIMENTAL DESIGN

#### A. Local paired study

We evaluate five synthetic streams, four controlled ML/agent patterns, and three instrumented algorithm traces. Every stochastic stream is materialized once per seed and shared byte-for-byte across configurations. Ideal uses lookahead  $D = 1$ ; predeclared timed and constrained points use  $D = 16$  and  $D = 8$ , with the full  $D \in \{1, 8, 16, 32\}$  sweep retained.

A SHA-256 identifies each stream. Repeated deterministic runs on one hash are one experimental unit rather than pseudo-replicates; stochastic streams and MLP seeds remain distinct. We report the paired cycle ratio

$$B(A, C) = \text{mean}_s C_s / A_s, \quad (2)$$

where  $B > 1$  favors candidate  $A$  over control  $C$ . Exact sign tests receive one Holm correction over the saved comparison family.

#### B. Frozen native holdout

We implement the gate as a ChampSim runtime L1D prefetcher and use official `develop commit e6530c7` [6]. One binary selects no L1D prefetch, official `ip_stride`, gate-disabled matched stride, or gated stride through JSON. The disabled module must match official `ip_stride` exactly on instructions, cycles, cache metrics, prefetch metrics, and DRAM reads/writes.

After an abbreviated eleven-program external development pilot, we freeze the nine SPEC CPU2017 programs not yet observed. Each uses its highest-weight public DPC-3 SimPoint [7]. The primary run warms 50 million and measures 200 million retired instructions. One program is one independent unit; SimPoint weights select a trace and are not replicate counts.

Before holdout, we predict that gating will (1) reduce aggregate accepted L1D prefetches and L2C/LLC prefetch

TABLE I  
MLP VS. REGULARITY-FILTERED STRIDE, LOCAL TIMED 8-WAY,  $D = 16$ .  
 $B > 1$  FAVORS THE MLP; HOLDOUT SEEDS 10–39.

| workload         | $B(\text{MLP}, \text{regstride})$ | Holm result |
|------------------|-----------------------------------|-------------|
| random           | 0.99990                           | n.s.        |
| phased           | 0.99476                           | MLP slower  |
| KV-cache append  | 1.10087                           | MLP faster  |
| embedding lookup | 0.87515                           | MLP slower  |
| agent RAG        | 0.94367                           | MLP slower  |
| strided read     | 0.51016                           | MLP slower  |
| mixed read/write | 0.12810                           | MLP slower  |
| token stream     | 0.94413                           | MLP slower  |

misses, (2) increase accepted-prefetch accuracy while retaining most useful prefetches, (3) leave DRAM reads within a frozen  $\pm 1\%$  descriptive margin, and (4) not improve geometric-mean IPC. We report a deterministic 10,000-resample program bootstrap and an exact two-sided sign test. The  $\pm 1\%$  rule is not a formal equivalence test.

ChampSim’s “requested” counter includes queue retries, so accepted L1D prefetches use “issued.” L2C/LLC prefetch misses show how much accepted activity descends the hierarchy; total DRAM reads are the off-chip endpoint.

### IV. RESULTS

#### A. Matched controls falsify broad neural superiority

Table I compares the MLP with regularity-filtered stride at timed  $D = 16$ . Random traffic is indistinguishable. The MLP has a conditional KV advantage but loses on most regular and mixed streams. No general neural lead survives the matched comparison.

The KV case suggests that a bounded regressor can preserve a local delta while ignoring gathers. It is a conditional mechanism, not evidence of superiority: the best classical configuration changes with profile and lookahead.

#### B. Local holdout suggests a useful gate

At constrained  $D = 8$ , regularity-filtered stride beats raw stride by  $1.08838\times$  on random,  $1.07587\times$  on phased, and  $1.01594\times$  on agent-RAG streams. Accepted-prefetch reductions are 99.55%, 83.07%, and 20.86%. Selected sequential, strided, mixed, embedding, and KV controls are unchanged. Timed  $D = 16$  gives  $1.03964\times$  on random and  $1.03298\times$  on phased traffic, each with 30/30 paired wins.

This is not a no-harm theorem. At ideal  $D = 1$  the gate slows agent RAG by 1.30%; on one deterministic merge-sort trace it is 5.95% slower. Larger lookahead reverses that trace. These counterexamples motivate native endpoint validation rather than a positive local conclusion.

#### C. Native holdout invalidates the proxy inference

The matched raw module equals official `ip_stride` on every recorded field and all nine programs: its paired IPC ratio is exactly 1 with nine ties. The gate comparison is therefore not an accidental reimplement change.

TABLE II  
FROZEN NATIVE CHAMPSIM HOLDOUT, GATE VS. MATCHED RAW STRIDE.  
REDUCTIONS ARE AGGREGATE COUNTS OVER NINE EQUAL-LENGTH  
PROGRAMS.

| metric                     | gate vs. raw stride |
|----------------------------|---------------------|
| geometric-mean IPC ratio   | 0.99434             |
| accepted L1D prefetches    | -36.77%             |
| L2C prefetch misses        | -7.33%              |
| LLC prefetch misses        | -6.85%              |
| DRAM read requests         | -0.083%             |
| accepted-prefetch accuracy | 10.64% → 15.62%     |
| useful-prefetch retention  | 92.83%              |

Table II gives the frozen endpoint result. The gate removes 36.77% of accepted L1D prefetches and improves accuracy by 4.98 percentage points while retaining 92.83% of useful prefetches. The apparent improvement shrinks to 7.33% and 6.85% fewer L2C and LLC prefetch misses and only 0.083% fewer DRAM reads. Raw stride is  $1.06691 \times$  no-prefetch IPC; gated stride is  $1.06087 \times$ . Direct gate/raw geometric-mean IPC is 0.99434 (bootstrap 95% CI 0.98445–0.999997).

The IPC loss is not directionally universal. GCC loses 4.37%; the other ratios range from 0.99626 to 1.00076. There are five losses and four wins, so the exact sign-test result is  $p = 1$ . The bootstrap interval reflects the magnitude of GCC; the sign test reflects inconsistent direction. By contrast, every program changes DRAM reads by at most 0.524%, within the frozen margin.

The correct interpretation is not “gating saves memory bandwidth.” Most suppressed prefetches either do not descend the hierarchy or return later as demand reads. Accuracy and L1D issue count are useful diagnostics, but they are not endpoint evidence.

## V. THREATS TO VALIDITY

The native study improves external validity but is not silicon. It uses one highest-weight SimPoint for each of nine holdout programs, not all SimPoints; one ChampSim configuration; and no multicore interference. SPEC CPU2017 does not validate the workload-shaped ML/agent streams. Storage accounting is not RTL synthesis and establishes no energy or critical-path result.

Internal validity is stronger. Local configurations share exact stream hashes and separate lateness, pollution, and drop metrics. Native configurations share one binary and trace; source, config, binary, and trace hashes are saved; raw stride matches official `ip_stride` exactly; and predictions were frozen before holdout.

Custom cache-callback counts differ by up to 0.105% on LBM even with identical retired-instruction counts. ChampSim changes phase at core retirement while requests remain in flight, so timing changes how many callbacks cross the boundary. We report this diagnostic but do not use it as an experimental unit or silently relax an equivalence test. Exact raw/official endpoint equivalence remains the hard invariant.

Future evidence should use all-SimPoint weighting, additional cache and multicore configurations, a second simulator, and hardware synthesis. The current work supports a negative/methodological architecture paper, not a deployable performance or energy claim.

## VI. REPRODUCIBILITY

The local artifact has 21,240 raw rows; the native artifact has 36 raw runs. Both record source, configuration, unit, and result provenance. Native metadata additionally pins the ChampSim revision and binary, module, trace-list, and trace hashes. Run:

```
python -m benchmarks.verify_research
python champsim/verify_results.py \
    --results \
    results/champsim_holdout_standard
```

The second command reconstructs every aggregate from raw JSON/logs, checks official-control equivalence, and evaluates the frozen predictions. Optional checkout and trace arguments verify external binary and trace hashes.

## VII. CONCLUSION

A matched control changes the interpretation of this learned-prefetching study. The MLP is neither generally safer nor generally faster; its persuasive random-stream behavior is reproduced by a classical predictor with the same admission policy. Native validation then exposes a second failure: a simple gate strongly improves accepted-prefetch count and accuracy, but DRAM reads do not materially change and mean IPC falls slightly on frozen holdout.

Admission remains a possible local activity/performance trade-off. It is not, on this evidence, an off-chip bandwidth optimization. The publishable result is the matched-control, cross-model falsification and its machine-verifiable demonstration that proxy metrics must not substitute for endpoints.

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